



Research papers

Influence of vegetation type and soil properties on soil water dynamics in the Šumava Mountains (Southern Bohemia)



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ABSTRACT

This study focuses on the description of soil water dynamics at four sites with different land cover types, namely beech forest, conifer forest, meadow and clipped grass. The analysis was based on soil tensiometer measurements from five consecutive vegetation seasons (comprising both wet and dry years). We investigated both column average pressure heads and also their vertical distribution. The soil water balance was studied by the HYDRUS-1D model.

The highest pressure heads were observed at the grassland site, followed by the meadow site. The forested sites were generally reaching lower pressure head values, which was a result of higher evapotranspiration and different soil properties. The differences between the spruce forest (*Picea abies* (L.)) and beech forest (*Fagus sylvatica* L.) were evident namely in dry periods, when the beech site was experiencing lower pressure heads. Contrarily, the spruce site was drier (with recorded lower pressure heads) in wet periods and at the beginning of each season. Compared to the conifer forest, lower pressure heads were observed in beech forest, namely at the bottom of the inspected soil column (down to 100 cm). The inspection of the soil water balance revealed different rates of evapotranspiration and drainage at all sites. The evapotranspiration was highest in the beech canopy followed by spruce and both grass covered sites. The differences between spruce and beech forest were based namely on the water consumption efficiency and differences in interception rates, vertical distribution of the roots, and soil hydraulic properties.

1. Introduction

An understanding of spatial and temporal variation of soil moisture is essential for studying other hydrological, biological or chemical soil processes, such as water movement, microbial activity and biogeochemical cycling (Ridolfi et al., 2003; Legates et al., 2011; Zarlenga et al., 2018; Gao et al., 2019). Although the world-wide total amount of water stored in the soil profile is negligible compared to ocean and glacier storages, it represents a crucial variable in terms of water resources and agricultural management. This is valid especially in the context of the ongoing shift in climate.

Soil water exhibits a tremendous heterogeneity both in space and time (Gomez-Plaza et al., 2000; Espejo-Pérez et al., 2016; Tauro et al., 2018). Therefore, spatial and temporal variations of soil moisture have always been a critical issue. The spatial variability is influenced by several factors, ranging from the topographical effect on lateral water redistribution (Williams et al., 2003), solar radiation (Geroy et al., 2011; Grayson et al., 1997), soil texture and structure (Famiglietti et al., 1998; Pan and Peters-Lidard, 2008), vegetation (Teuling and Troch, 2005),

climate (Lawrence and Hornberger, 2007), precipitation patterns (Keim et al., 2005) and antecedent soil moisture (Rosenbaum et al., 2012).

The land cover and land use influence on the soil water regime is complex involving various processes like interception and transpiration (Sutanto et al., 2012; Jakobi et al., 2018). In addition, vegetation can also influence the soil hydraulic properties, i.e. hydraulic conductivity and water retention characteristics (Haghighi et al., 2010; Zhou et al., 2008). Several previous studies have investigated spatial and temporal variation of soil water under a certain land cover type (Gross et al., 2008; Zhou et al., 2008). Nevertheless, studies comparing the influence of several land cover types on soil water regime over the period longer than one or two years (studied e.g. by Huang et al., 2016; Korres et al., 2015; Zucco et al., 2014) are missing. Thus, there is still a need to compare the dynamics of soil water regime under different land use types and longer time periods (Niu et al., 2015).

This knowledge would be valuable especially in the context of extreme climatic events that occur nowadays. In central Europe, a period of major floods (1997, 2002, 2013) was followed by serious dry spells (2003, 2011–2012, 2015) (Trnka et al., 2015). This observed

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hydrological extremity raised the question of a sustainable water management. One of the possible management practices in consideration is represented by the land cover changes intended to hold more water in the landscape and simultaneously to attenuate the rainfall-runoff response.

The aim of this study is therefore to understand the soil water regime based on soil tensiometer measurements in the vegetation season under four different land covers (coniferous/deciduous forest, meadow, grassland). Another objective is to describe possible differences in the water balance, which strongly influences the soil water regime during the vegetation season. This could help to explain the soil water behaviour in the region when exposed to a water deficiency or abundance. This analysis was therefore conducted in five consecutive years, encompassing both dry and wet periods.

2. Data and methods

2.1. Study site

The *Liz experimental catchment*, Czech Republic (49°04'N, 13°41'E) was used as an experimental area for this study. The catchment lies in the in the Šumava Mountains (Southern Bohemia), which form a boundary between the Czech Republic and Germany. It belongs to forested mountainous region, which serves as a headwater area for numerous flood events. The soil type is classified as moderately deep loamy sand dystric Cambisol. The average soil depth is estimated to be approximately 100 cm. The soil in the area generally has high water permeability (518–1700 cm.day⁻¹ from the bottom to the top of the soil profile). The saturated hydraulic conductivities were measured using a pressure infiltrometer (Pavelková, 2003). The geological base is represented by crystalline hard rocks, which enable water circulation only in the weathered zone (Hrkál et al., 2009). Hence, in the Liz experimental area, the hydrological catchments correspond to the hydrogeological ones. The majority of the area is covered by mixed forest, with prevailing coniferous trees (*Picea abies* (L.)) of various ages (up to 140 years). According to Köppen climate classification the site lies in the Dfb climate zone characterised by a humid continental climate with approximately uniform precipitation distribution and warm summers. The average annual temperature is 6.6 °C. The average annual precipitation is approximately 863 mm. A more detailed description of the site is available in Šípek and Tesar (2016).

2.2. Field measurements

The experimental site was equipped with an automatic system for measuring meteorological and hydrological variables via wireless data transmission. The fundamental meteorological measurements necessary for this study comprise air temperature and relative humidity (Fiedler RV12/RK5, Czech Republic) and precipitation (Meteoservis MRW 500, Czech Republic). The meteorological data were measured at a 10-minute interval during the entire period. The potential evapotranspiration was estimated using the method introduced by Priestley and Taylor (1972). All the necessary variables were available. The only exception was the net longwave radiation, which was estimated using the FAO56 approach with the site specific coefficient values (Kofronova et al., 2019).

The soil water regime was measured by means of permanently installed soil tensiometers (Adolf Thies GmbH, Germany, Supplementary material Fig. S1). The sampling strategy was set up in order to comprise sites with different land cover and simultaneously with similar meteorological conditions. Hence, the soil water pressure heads were measured at four locations, representing four different land covers (clipped grassland – GRA, meadow – MEA, spruce forest – SPR and beech forest – BEE). The maximum distance between the sites was 1355 m. The location of particular sites is shown in Fig. 1, and their overall characteristics are given in Table 1. At each site, the pressure heads were measured at five depths (15, 30, 45, 60 and 90 cm). The measuring range of these tensiometers accounted for the pressure heads

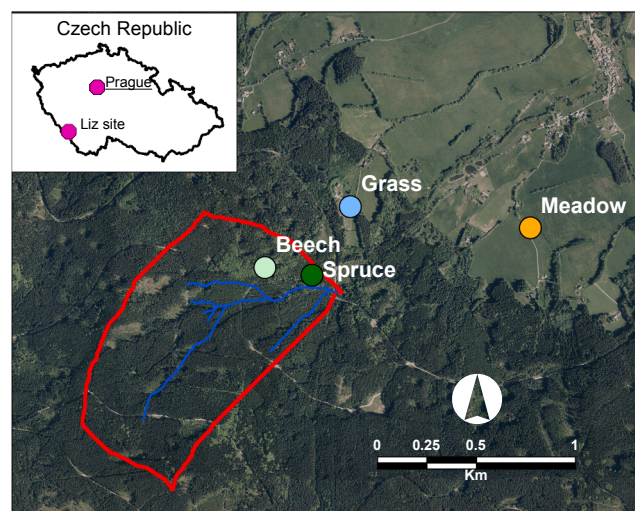


Fig. 1. Location of the experimental site. Colours of particular sites are the same throughout the entire manuscript.

Table 1

Experimental sites characteristics. The parameters of the soil water retention curve denote: residual water content – θ_r , water content at saturation – θ_s and two shape parameters α and n . MEA stands for meadow, GRA for clipped grassland, SPR for spruce forest and BEE for beech forest.

	MEA	GRA	SPR	BEE
Land cover	Meadow	Grass	Spruce	Beech
Sand [%]	57	49	73	80
Silt [%]	36	46	24	18
Clay [%]	7	5	3	2
θ_r	0.174	0.009	0.193	0.154
θ_s	0.387	0.424	0.523	0.461
α	0.063	0.036	0.055	0.051
n	1.275	1.128	1.652	1.462
RMSE [%]	1.62	0.95	2.65	0.47

from 0 cm to –867 cm (–85 kPa). The pressure head values were recorded manually three times a week during the vegetation season (mid-May to mid-October) from 2003 to 2007. One to four tensiometers were available for each measuring depth at each site, and the single value for a particular depth was taken as their average. The median standard deviation of daily measurements (taking into account all measurements for each depth) ranged between 5 and 15% of the observed mean daily value. It was generally highest in spruce forest and dry periods. Moreover, in all land covers, it decreased with depth. Average soil column pressure head was estimated as a weighted mean of five soil layers (each being represented by one measurement depth). The soil layers were separated by the depths of 22.5, 37.5, 52.5 and 75 cm (lying exactly between the two adjacent tensiometer measurement points). The soil profile was considered to be of the uniform depth of 100 cm. The quality of the available tensiometric measurements (Adolf Thies GmbH, Germany) was compared with widely used UMS T8 tensiometers (UMS, Germany) at the SPR site (UMS T8 were installed in 2009). The pressure heads measured by both tensiometer types corresponded to each other satisfactorily over the following five year period, which is shown in the Supplementary material (Fig. S2).

2.3. Climate and soil characteristics

The climatic characteristics of the study period are given in Fig. 2. Each year was split into a cold (November–April) and a warm (May–October) season. The five evaluated years represented very different climatic conditions when compared to long-term average values.

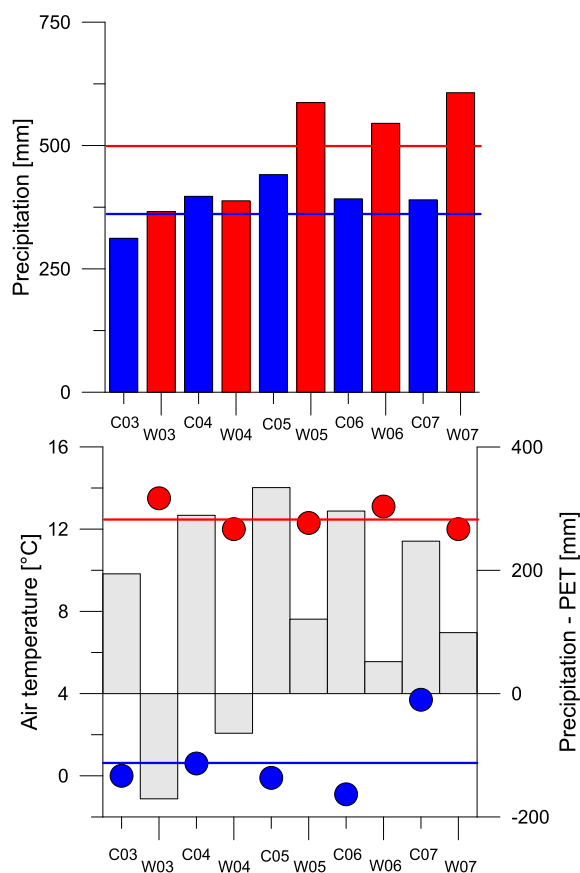


Fig. 2. Climate characteristics of cold (blue colour, C03–C07) and warm (red colour, W03–W07) seasons in the period of 2003–2007. The bar plot (upper panel) represents an actual sum of precipitation. The scatter plot (lower panel) depicts an average daily air temperature in each season and the grey bar plot (lower panel) denotes the difference between the sum of precipitation and potential evapotranspiration. The lines of corresponding colour stand for long-term average (1976–2015) sum of precipitation and air temperature (for both cold and warm seasons separately). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The year 2003 was extremely dry, with above average temperatures in the warm season. The sum precipitation in the warm season was lower than potential evapotranspiration by 120 mm. The year 2004 was also dry, but the preceding winter was wetter than on average. Hence, the initial wetness was substantially higher compared to 2003. Initial pressure heads were in the range from -70 to -100 cm, compared to -120 to -280 cm in 2003. The sum of precipitation was again lower than potential evapotranspiration but the difference decreased to 41 mm. The wettest year observed was 2005, having slightly below average temperatures and above average precipitation both in the warm and the cold seasons. The very wet antecedent winter was reflected by the highest initial wetness of all inspected years (2003–2007) ranging from -50 cm (Grass site) to -110 cm (Spruce site). The subsequent year, 2006, was also wetter than average, but compared to 2005, the temperatures were lower during the cold season and higher in the warm season. The snow cover in this winter was the highest observed since the beginning of the observations in 2002. The average snow water equivalent over the 2005/2006 winter season ranged from 90 to 120 mm (depending on the location), compared to the long-term average of 40–65 mm (Šípek and Tesar, 2014). The last year of the experiment (2007) was again very wet. This was reflected by the excess of precipitation over the potential evapotranspiration attaining 166 mm. However, the preceding winter (2006/2007) was the warmest winter observed, with nearly no snow cover. This significantly influenced the pressure heads in the warm season.

Undisturbed soil samples (collected in stainless steel sample rings of a high of 6 cm and volume of 138 cm^3) were used to determine the soil water retention curves (SWRC) for each site and depth. A number (from two to six) of measured SWRCs were available for each depth and site. The sand table method was used for pressure heads in the range from -1 to -30 cm. The pressure plate extractor (Soilmoisture Equipment Corp., USA) was used for the pressure head range from -102 to -15296 cm (10 to 1500 kPa). The volumetric soil moisture content was determined for 7–11 specific pressure head values. The SWRC were fitted by means of the RETC software (van Genuchten et al., 1991) using the Van Genuchten (1980) equation. All SWRCs are present in the Supplementary material (see Fig. S3). The column average Van Genuchten (1980) SWRC parameters are shown in Table 1. Generally, the forested sites were drained more easily and had a higher saturation water content, which was determined by the higher portion of sand particles in the soil column.

In this study, the primary analyses were conducted using the pressure head data obtained directly from the soil tensiometers. The reason for that was the uncertainty related to the conversion of pressure heads to volumetric water contents. This uncertainty is based on the dependence of the equilibrium soil water content on the direction of the soil wetting/drying process (Hillel, 2003). For comparability of the results with other studies, the information about volumetric water contents obtained by the soil water retention curves from the tensiometer data is presented in the Supplementary material (Figs. S5–S7).

2.4. Statistical analysis

The differences between the means of pressure heads from individual sites and depths were statistically assessed. Analysis of variance (ANOVA) is often used for testing the equality of means; nevertheless, the crucial assumption of ANOVA is the equality of variances of individual samples. This assumption was tested using the Brown and Forsythe test (Brown and Forsythe, 1974). In all cases, the Brown and Forsythe test confirmed the significant differences between the variances of individual datasets; therefore, the ANOVA was not considered and the non-parametric Kruskal-Wallis test (Kruskal and Wallis, 1952) was applied instead. The Kruskal-Wallis test examines the hypothesis that the individual samples originate from the identical distribution; nevertheless, it is particularly sensitive to differences in the means. All tests presented in Section 3 were performed at a significance level of 5%.

2.5. Water balance modelling

2.5.1. Soil water model

The HYDRUS-1D model is a standard tool for the simulation of water and solute movements in unsaturated, partly saturated or fully saturated porous media (Simunek et al., 2008). The HYDRUS-1D model designed for one-dimensional fluxes is based on the modified Richard's equation with the assumption that the air phase interactions with the liquid flow process might be neglected, and water flow due to thermal gradients is also insignificant. The model version using a single porosity hydraulic model was used. This was based on several extensive side-experiments in which different hydraulic models were examined and the single porosity model was generally the most efficient one (reaching lowest RMSE). The unsaturated soil hydraulic properties were described using Van Genuchten (1980) type of analytical functions. In HYDRUS-1D model, the actual root water uptake (RWU) was represented by Feddes equation (Feddes et al., 1974). Beer's law method was used to partition potential transpiration and soil evaporation fluxes based on the leaf area index (LAI).

The depth of the modelled soil profile was set to 100 cm because the soil–bedrock interface takes place at this depth. The soil profile was divided into five distinct layers ($0-22.5-37.5-52.5-75-100$ cm). For each layer, the tensiometer measurement was available (see Section 2.2) and used for the model calibration. The lower boundary condition

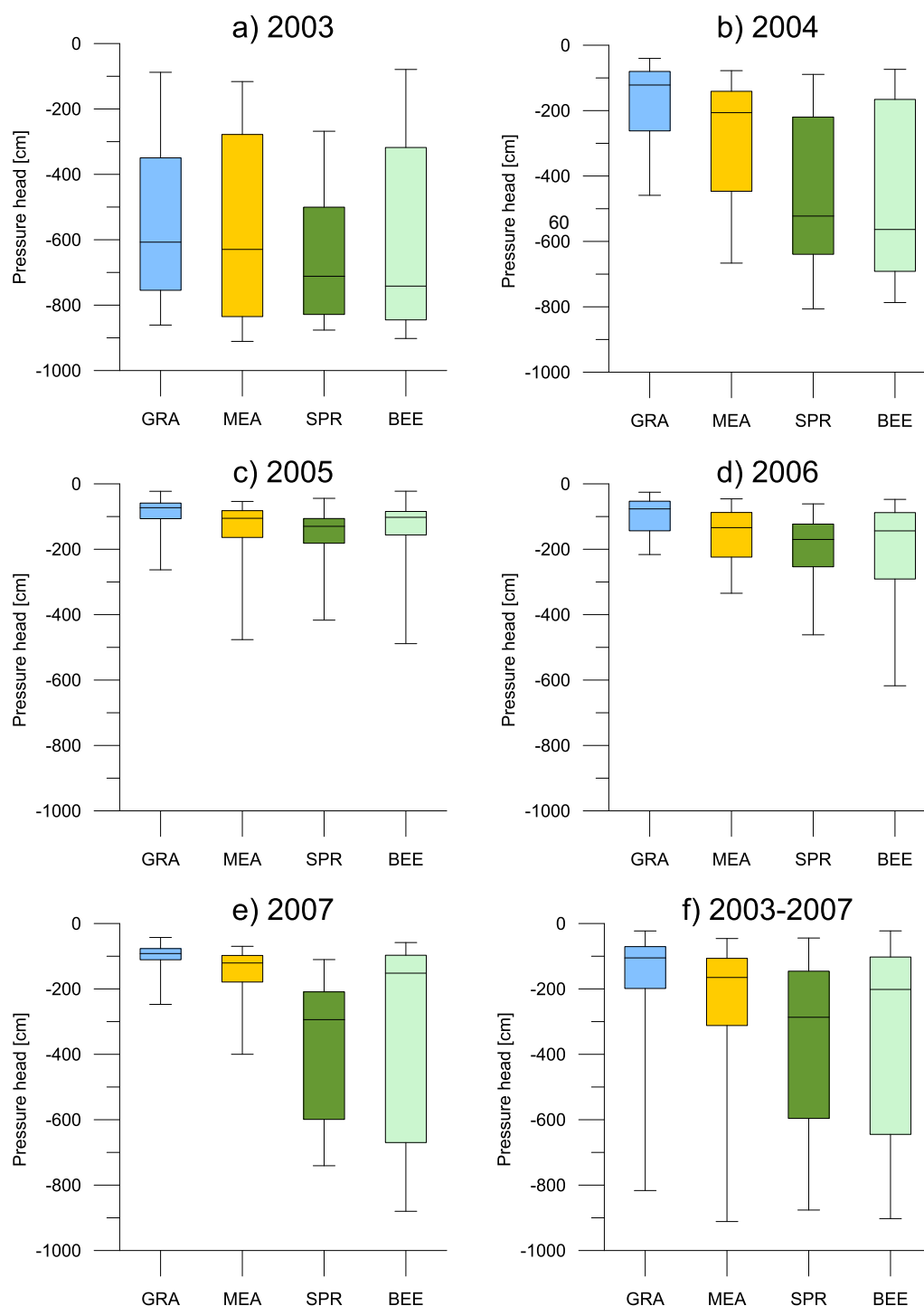


Fig. 3. Box plots of all column average pressure heads observed in each individual year and site (a–e). The last plot (f) depicts all observed pressure head values over the entire five-year period.

was represented by free drainage. The atmospheric boundary condition with surface runoff was used as an upper boundary condition. The grass interception capacity was obtained based on side experiments in HYDRUS-1D comparing several scenarios, where the interception storages ranged between 0 and 2 mm. The value of 1.1 mm was chosen to be the most efficient. It also fits the observed range by [Thurow et al. \(1987\)](#) investigating the interception characteristics in grasslands. The LAI of both grass covered sites was set to the value of $2.4 \text{ m}^2 \cdot \text{m}^{-2}$, which was estimated based on FAO56 approach ([Allen et al., 1998](#)). The interception capacity of the spruce covered site was set to 2 mm, which was reported by [Dohnal et al. \(2014\)](#) based on the measurement using five

raingauges (measuring throughfall) located in the same experimental site. In the same study, the measured average LAI of spruce equalled to $5.83 \text{ m}^2 \cdot \text{m}^{-2}$. The interception capacity of the beech site was set to 1.5 mm, which was obtained from a similar experiment estimating throughfall in beech canopy at the BEE site. The LAI of the beech site was set seasonally variable (ranging from 1.27 to $5.08 \text{ m}^2 \cdot \text{m}^{-2}$ from May to October). This range was reported by [Haninec \(2014\)](#), who compared several methods of LAI estimation in the beech forest in the Czech Republic. The depths of the rooting zones were not exactly quantified. However, based on the visual inspection (when excavating the soil) they were set to 40 cm (GRA, MEA and SPR sites) and to

100 cm (BEE site). This is in agreement with Schmid (2002) and Schwärzel et al. (2009) who reported similar root distributions in the spruce (down to 40–50 cm) and beech stands (down to the depth of 80–90 cm) in Cambisol sites in Austria and Germany.

2.5.2. Modelling procedure

The soil water model was used to optimise the root water uptake parameters for each particular place separately. For this experiment, the measured soil water retention curves were utilised and saturated hydraulic conductivities (K_{sat}) were inversely estimated (the measured K_{sat} was available only at the SPR site). The parameters of the root water uptake model were manually fitted (Hydrus-1D does not enable their automatic optimisation) so that the modelled soil water regime matched closely to the observed one. The objective criteria for this match composed of RMSEs and visual inspection of the resulting plots. The parameters that were adjusted consisted of threshold values of the Feddes et al. (1974) function denoting the pressure head below which roots start to extract water from the soil (P_0), the pressure head below which roots extract water at the maximum possible rate (P_1), value of the limiting pressure head, below which roots can no longer extract water at the maximum rate (P_2), and value of the pressure head, below which root water uptake ceases – usually taken as the wilting point (P_3).

3. Results

3.1. Measured soil water regime in different land cover types

The measured pressure heads were analysed from two different points of view. The first was focused on the column average pressure heads, while the second was represented by the vertical distribution of the pressure heads in the soil column (down to the depth of 100 cm). For clarity, the pressure heads from individual sites/seasons/depths are presented as box plots.

3.1.1. Measured average pressure head in the entire soil column

All observed pressure heads for each site and year separately are shown in Fig. 3. The values were season dependent and might be roughly divided into wet and dry years. In the dry years 2003 and 2004, the beech site was attaining lowest pressure heads (the median observed values were -751 and -534 cm). The median pressure heads in the spruce forest (SPR) were higher by 21 and 23 cm, respectively. The pressure heads in the spruce forest site were lower than the beech site (BEE) only when the observed precipitation was average or above-average (2005–2007). Moreover, the differences in pressure heads between spruce and beech plot were actually more pronounced as the tensiometer data often reached the lower measurement limit of -867 cm in some depths especially in the beech forest. This was the case of the year 2004 and 2007 (Fig. 4), when the pressure heads in beech forest were most probably lower than those recorded by the soil tensiometers (see Fig. S4 – the reading from the depth of 90 cm). The entire soil profile was affected by the measurement restriction in 2007 (not only at a depth of 90 cm measurement). In the year 2003, the pressure heads at all sites exceeded the measuring range (0 to -867 cm). Hence, the differences in the lowest pressure heads were biased. Nevertheless, in the case of pressure heads lower than the measurement limit, the loss/gain of the volumetric water content corresponding to the unit change in pressure head is very small. It accounts for less than $0.002 \text{ cm}^3 \cdot \text{cm}^{-3}$ of change in the volumetric water content per a 100 cm change in pressure head. The same rate observed from the saturation to the pressure head of -100 cm equals to $0.22 \text{ cm}^3 \cdot \text{cm}^{-3}$. Hence, the changes in pressure head concerning such low heads result in a very little change in the amount of water present in the soil profile and simultaneously very little variation in water fluxes. The differences in pressure heads between beech and spruce sites were statistically significant only in the dry year 2007 (Fig. 5).

The differences between spruce and beech sites were more significant in terms of volumetric water contents (Figs. S5 and S6). The spruce site was significantly wetter in dry years. In wet years, the differences were of lower magnitude but of the same character as in pressure head datasets. The reason is that the soil water retention curves representing these two sites do not intersect and the particular pressure head always implied higher volumetric soil water content in the case of the spruce site compared to the beech site.

The pressured heads and volumetric water contents in grassland (GRA) and meadow (MEA) sites were generally higher than in the forested sites (SPR&BEE). Except for the year 2003, the interquartile range of pressure heads (Q_2 – Q_3) was higher in the forested sites and lower in meadow and grassland sites. These observations were confirmed by the results of the Kruskal-Wallis test (Fig. 5), showing prevailing significant differences among GRA/MEA and forested sites.

3.1.2. Vertical distribution of the pressure heads

The box plots depicting the vertical distribution of all pressure head values in the seasons with contrasting precipitation, 2004 and 2005, are shown in Fig. 6. Moreover, the four-year temporal evolution of pressure heads with respect to depth is depicted in Fig. 4. These years were selected as examples of dry and wet seasons. In drier years, the beech and spruce forests slightly differed in their vertical distribution of pressure heads. The beech forest extracted water more efficiently from higher depths, which resulted in a higher variation of pressure heads. Contrarily, in the topsoil (depth of 0–37 cm), median pressure head values were slightly lower in the spruce forest. This is nicely shown on the evolution of the pressure heads from the beginning of August to the end of September in 2004 (Fig. 4). Fig. 4 (namely the years 2003, 2004 and 2007) also documents the drier soil under spruce forest at the beginning of the season, which can be attributed to higher interception of conifer forest compared to deciduous one. Moreover, it also shows that GRA/MEA/SPR sites dries gradually from the soil surface towards the deeper layers. Contrarily, at BEE site the lower pressure heads were usually observed first in the deeper layers. However, the Kruskal-Wallis test did not confirm any significantly different pressure heads between SPR and BEE (Fig. 5). On the other hand, the statistically significant differences were found in the vertical distribution of volumetric water contents (Fig. S6). In wet years, the differences between spruce and beech sites were even less obvious and the vertical distribution of pressure heads was similar to those of grassland and meadow sites (see year 2006 in Fig. 4). This underpins the fact that when water is abundant, the differences in pressure heads among particular land covers are of low importance.

In wet seasons, the pressure heads in GRA/MEA/SPR/BEE sites can be characterised by an approximately uniform vertical distribution, rarely attaining values lower than -200 cm. At all four sites, the pressure heads were increasing with increasing depth (Fig. 6). The statistically significant differences among GRA/MEA and forested sites were regularly identified at the bottom of the soil profile (below 45 cm). However, they were based on narrow distributions of pressure heads. In the topsoil, the significant differences concerning GRA/MEA and forested sites were restricted to dry years (Fig. 5).

3.2. Soil water balance modelling

The measured pressure heads were used to calibrate the HYDRUS-1D soil water model for all four inspected sites. The main objective was to adjust the root water uptake (RWU) model parameters so that the soil water regime simulations would be as efficient as possible. The model results are used to differentiate particular water fluxes at all sites in all seasons. Namely, the rates of evaporation from soil column and canopy, plant transpiration and drainage are of particular interest.

The root water uptake model parameters were identified for each site separately (Table 2), representing a specific land cover characteristics. Resulting model efficiencies in terms of RMSE, coefficient of

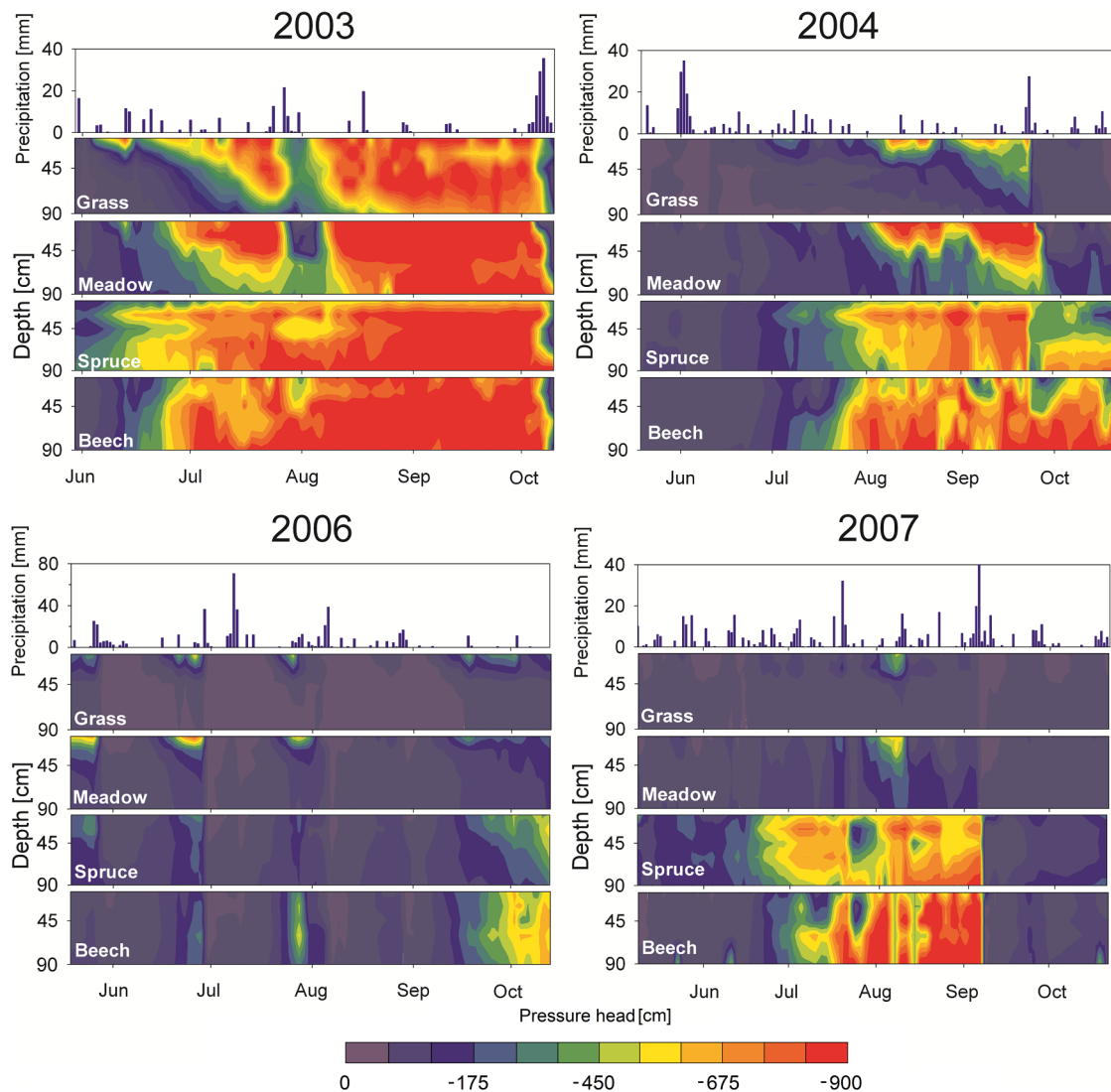


Fig. 4. Daily open area precipitation and the development of pressure heads at all sites for four selected vegetation seasons.

determination (R^2), and Nash-Sutcliffe coefficient are shown in Table 3. The estimated pressure heads varied from the observed ones most pronouncedly in both forested areas (RMSE range extending from 100 cm to 176 cm). The grass covered sites were generally better represented by the model (with the exception of the year 2003). The reasons for such high error statistics are further discussed in the Discussion section.

The estimated water balances are for each year depicted in Table 4. In every single year the beech site had the highest amount of actual evapotranspiration – AET (sum of canopy transpiration, evaporation from soil surface and interception loss). The differences were from 62 to 81 mm per season compared to the SPR site and from 44 to 105 mm per season compared to grass covered sites (GRA/MEA). The differences between particular sites followed the same pattern every year, only their absolute values were variable.

The highest transpiration rates were always observed in beech forest (seasonal average equalled to 278 mm over the five year period), followed by spruce one (230 mm) and lowest rates were in the grass covered sites (164 and 173 mm). As shown in Fig. 7, the differences in transpiration between beech and spruce trees were usually increasing from the beginning to the middle of vegetation season (when beech transpired at higher rate), which was followed by their gradual attenuation (higher transpiration of spruce compared to beech at the end of the season). This seasonality was one of the causes of significantly

lower pressure heads observed in summer in beech forest, albeit the annual differences were less pronounced.

Contrarily, the evaporation from the soil surface was highest in the grass covered sites with the season average of 144 mm in the GRA site and 127 mm in the MEA site, respectively. Both forest covered sites had lower rates of surface evaporation (~ 30 mm in spruce and ~ 75 mm in beech covered site, respectively). The different rates of evaporation can be explained by differences in the LAI and available soil water (see Table 2). The seasonal interception loss in particular sites corresponded to maximum interception capacity reaching highest seasonal averages in the spruce forest (100 mm per season). Finally, the drainage was highest in the spruce forest compared to all remaining sites. Altogether, the highest average annual evapotranspiration rate was observed in the beech forest ($431 \text{ mm} \cdot \text{year}^{-1}$) compared to all other remaining sites ($361\text{--}368 \text{ mm} \cdot \text{year}^{-1}$). The difference between spruce and grassland sites was namely in the above mentioned higher drainage in the spruce site ($89 \text{ mm} \cdot \text{year}^{-1}$ compared to 50 and 56 mm per a year).

4. Discussion

4.1. Measured pressure heads under different land covers

The spruce site was generally wetter compared to the beech site. This is in contradiction with the previous findings of Schume et al.

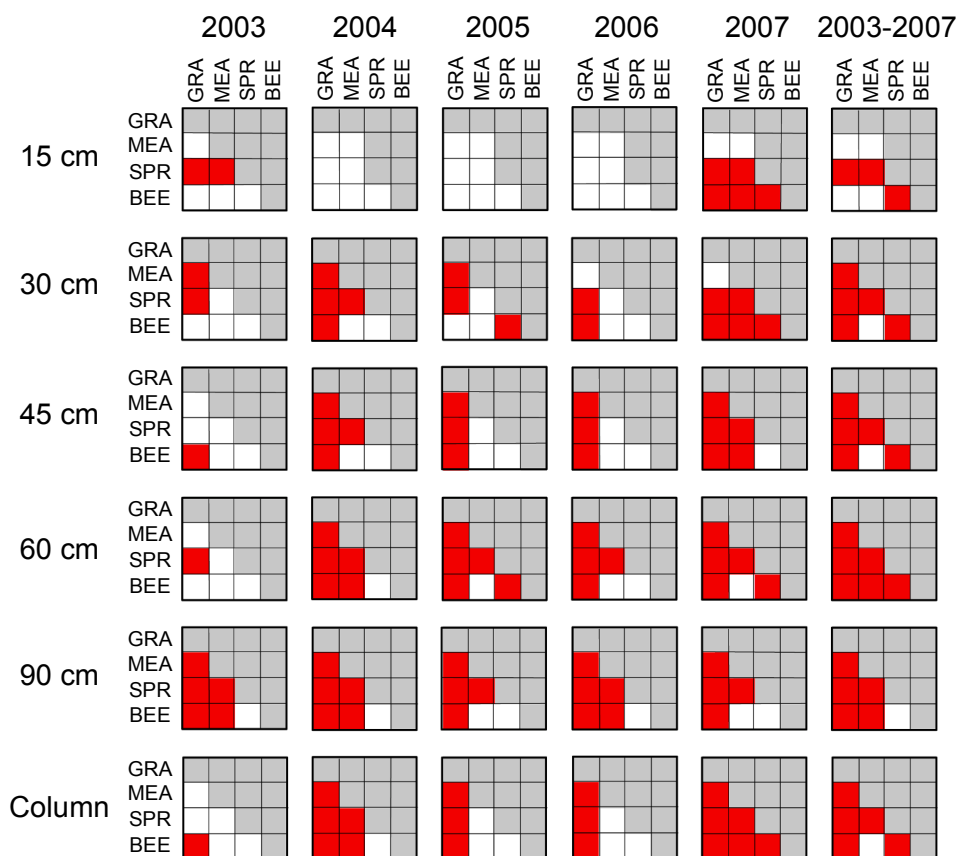


Fig. 5. Results of the Kruskal-Wallis test across all sites, years, and depths using pressure head values. Red squares indicate significantly different means (at the 5% significance level) and white squares indicate not confirmed differences in means. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(2004) and Jost et al. (2004), who observed wetter beech forest sites compared to spruce-dominated sites. Nevertheless, the comparison is influenced by the different soil properties. Both studies were conducted in the site with significantly higher clay content ranging from 37% to 56%, compared to 2–7% found in the Liz site. Besides the soil characteristics, the differences between spruce and beech forest could also be based on more economic transpiration demand (Cermak et al., 1995; Schume et al., 2004) and higher interception in the case of spruce (Augusto et al., 2002; Šípek and Tesar, 2014). Moreover, the midday stomatal closure (hence increased stomatal resistance and reduced transpiration) was observed in spruce forest by Zweifel et al. (2002) when the vapour pressure deficit became sufficiently high. Additionally, at the beginning of the vegetation season, slightly lower pressure heads were regularly observed in the spruce stand (Fig. 4). This was caused by the higher interception of the snowfall in the conifer forest observed in the area (Šípek and Tesar, 2014) and by the transpiration of spruce on favourable winter days (Boyce and Lucero, 1999; Mayr et al., 2002). Drier forest sites, compared to grassland and meadow ones, have been reported by several authors (Wiekenkamp et al., 2016a; Zehe et al., 2010; Zusco et al., 2014). This was caused mainly by the higher evapotranspiration of forested sites (as documented by the model in Fig. 7), which led to more pronounced declines in pressure heads.

The major difference in pressure heads depth distribution was caused by the different distribution of roots in Norway spruce (*Picea abies* (L.)) and European beech (*Fagus sylvatica* L.) forests (Schmid, 2002). While the fine roots of the spruce are mainly located in the topsoil (in a dystric Cambisol), the beech roots are distributed more uniformly. Similar character of root water uptake depth distribution was also reported by Schwärzel et al. (2009) in the Ore Mts., Germany.

As this study is based on a five-year period (covering both wet and dry years), we would like to stress that the prevailing character of the differences in pressure heads among particular land covers was partly

determined by the climatic characteristics of the particular season (as well as those of the preceding season). The partition into wet and dry years (according to precipitation amounts) was not sufficient, as the year 2007 was very wet in the vegetation season, but preceded by a relatively warm winter. The resulting soil water regime in 2007 thus rather corresponded to that of a dry year. Hence, to obtain representative results, it is necessary to conduct comparison studies on longer time scales. Short-term experiments bias the results of several studies and cause differences when comparing the outcomes. This could be partly the case of this study and the outcomes of Jost et al. (2004) and Schume et al. (2004), who documented drier spruce sites compared to beech sites (albeit only in two vegetation seasons, of which one was drier). However as mentioned above, they have investigated a site with different soil properties (namely significantly higher clay content ranging from 36% to 56%).

4.2. Water balance modelling

The soil water model root water uptake parameters were manually optimised in order to obtain most efficient simulation of the pressure heads in particular depths. The resulting parameters for the spruce site were in the range reported by Rabbel et al. (2018) and Vogel et al. (2013), who also focused on the RWU parameters in the spruce covered sites. The parameters for the grass covered sites differed only slightly from the default HYDRUS-1D parameters for the grass cover. The only difference between MEA and GRA sites was that the MEA site (where grass was not clipped so often) had a lower pressure head threshold denoting reduced actual evapotranspiration from its potential (P2), which could be explained by higher amounts of above ground biomass. The beech site model parameters obtained by the calibration procedure indicated namely the higher ability of the beech stand to extract water from deeper soil layers (Köstner, 2001). The most problematic was the wilting point (P3), which was in our study optimised to be around the

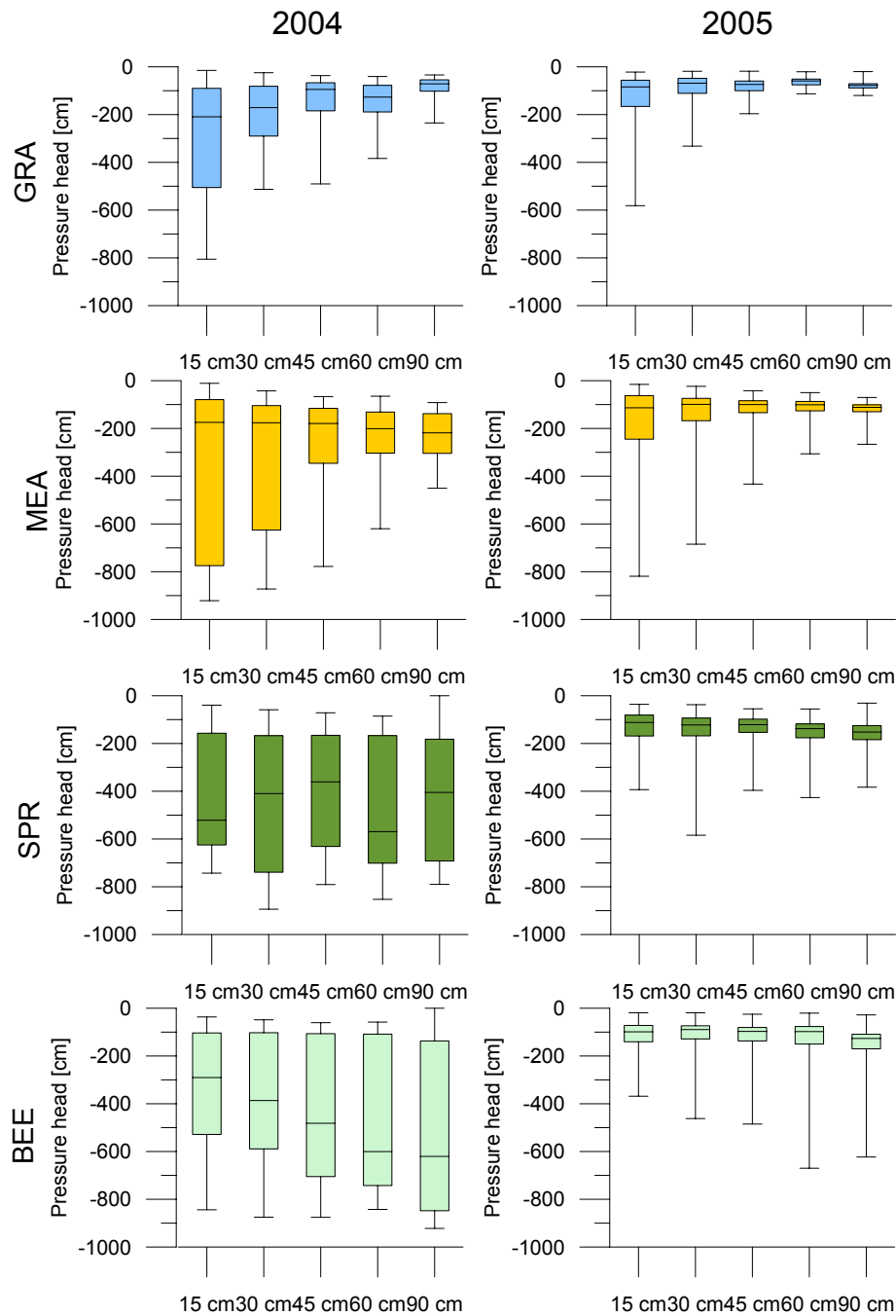


Fig. 6. Box plots depicting the vertical distribution of observed pressure heads in the dry (2004) and wet (2005) years for all inspected land cover types and measuring depths (horizontal axis).

Table 2

Site specific parameters of the HYDRUS-1D model governing the influence of land cover on the modelled soil water regime. MEA stands for meadow, GRA for clipped grassland, SPR for spruce forest and BEE for beech forest.

	MEA	GRA	SPR	BEE
Interception [mm]	1.1	1.1	2	1.5
LAI [$\text{m}^2 \cdot \text{m}^{-2}$]	2.45	2.45	5.83	1.27–5.08
P0 [cm]	–10	–10	–10	–10
P1 [cm]	–25	–25	–20	–10
P2 [cm]	–200	–100	–450	–800
P3 [cm]	–900	–650	–850	–900
Root depth [cm]	40	40	40	100

measuring limit of the tensiometer (Table 2). Hence, it does not represent the wilting point at all. The reason for this is twofold. First, when using the soil tensiometer, no reliable information is available about the course of soil water regime below its measuring limit. Therefore it is impossible to adjust the wilting point correctly. Second, the previously published shortcoming of the Feddes et al. (1974) root water uptake model by Vogel et al. (2013) showed incorrectly low pressure heads estimated by this approach compared to newly designed water potential based model. The larger declines in pressure heads were given by the late start of the reduction of transpiration (after the onset of dry conditions) and the permanent proportionality of root water uptake to the root distribution function. Hence in our study, the value of the “wilting point” (P3) does not correspond to its physical meaning but it is used only as a tool to capture the soil water regime as efficiently

Table 3

Hydrus-1D model efficiency in all sites and years. Only the average results for the entire soil profile are displayed. MEA stands for meadow, GRA for clipped grassland, SPR for spruce forest and BEE for beech forest.

		MEA	GRA	SPR	BEE
2003	RMSE [cm]	193	153	171	130
	R ² [-]	0.54	0.71	0.34	0.83
	NS [-]	0.76	0.87	0.75	0.88
2004	RMSE [cm]	151	53	144	162
	R ² [-]	0.78	0.80	0.21	0.78
	NS [-]	0.37	0.76	0.63	0.68
2005	RMSE [cm]	77	89	121	99
	R ² [-]	0.77	0.77	0.48	0.81
	NS [-]	0.73	0.38	0.71	0.82
2006	RMSE [cm]	77	47	163	157
	R ² [-]	0.52	0.67	0.63	0.60
	NS [-]	0.62	0.76	0.32	0.46
2007	RMSE [cm]	93	101	148	176
	R ² [-]	0.53	0.40	0.46	0.81
	NS [-]	0.51	0.09	0.44	0.63

Table 4

Annual water balance components [mm] for all inspected sites (2003–2005). MEA stands for meadow, GRA for clipped grassland, SPR for spruce forest and BEE for beech forest.

		SPR	BEE	GRA	MEA
2003	Precipitation	330			
	Transpiration	155	190	110	114
	Evaporation	27	74	115	108
	Interception loss	77	61	47	47
	Total actual ET	259	325	272	269
	Drainage	14	5	0	24
2004	Precipitation	328			
	Transpiration	188	242	106	132
	Evaporation	29	67	116	100
	Interception loss	95	76	58	58
	Total actual ET	313	385	281	290
	Drainage	68	7	0	53
2005	Precipitation	531			
	Transpiration	268	312	192	197
	Evaporation	33	72	154	124
	Interception loss	92	72	54	54
	Total actual ET	393	456	400	376
	Drainage	170	51	127	149
2006	Precipitation	523			
	Transpiration	277	336	223	214
	Evaporation	33	76	162	153
	Interception loss	97	76	59	59
	Total actual ET	407	488	444	426
	Drainage	141	105	87	53
2007	Precipitation	504			
	Transpiration	263	309	189	209
	Evaporation	38	85	172	151
	Interception loss	140	109	83	83
	Total actual ET	441	503	445	444
	Drainage	51	38	37	0
2003–2007	Precipitation	443			
	Transpiration	230	278	164	173
	Evaporation	32	75	144	127
	Interception loss	100	79	60	60
	Total actual ET	363	431	368	361
	Drainage	89	41	50	56

as possible – in order to obtain the information about soil column water balance.

Generally, the simulated pressure heads at the SPR site were slightly less efficient than those obtained by Vogel et al. (2013) focused on the same site but using a slightly different S1D model also based on numerical

solution of Richards' equation (Vogel et al., 2010) for a five month period. The efficiency of the pressure heads modelling was influenced by several factors that are not included in the modelling procedure. First, the rock fragments (their volume was not quantified in this study) could have influenced the estimate of the total amount of water present in the soil profile. HYDRUS-1D model generally assumes uniform distribution of water in each soil layer. However, if a certain percentage of the profile is formed by the rock fragments then the vegetation will extract more water from the areas between those rock fragments in order to fulfil the water demand. This could result in a higher actual drop in observed pressure heads (Vlček et al., 2012; Hlaváčiková et al., 2015), which would not be represented in the model. Second, the hydrophobicity of the soils (Doerr et al., 2000) may result in non-uniform drainage of water into deeper soil layers and formation of a shallow biomat flow (Gerke et al., 2015). The percolation of water can then be limited only to certain locations (eventually bypassing the measurement probes). Further, Wiekenkamp et al. (2016b) showed that the occurrence of preferential flow in the forested catchment can cause non-sequential reaction of soil moisture sensors at different depths. Micro-relief of the site could also bias the pressure head estimation in the same way due to the concentration of water infiltration into the small depressions as the surface is not completely flat in the SPR and BEE sites. Moreover, in these depressions the infiltration rate can be gradually reduced by clogging of soil pores by finer particles originating from the surface (Rousseva et al., 2002; Thompson et al., 2010). Finally, the course of pressure heads below the tensiometer detection limit (discussed in Section 2.2) is not known, hence the simulation of these low pressure heads was omitted by restricting the RWU limit to this threshold (P3) at maximum. However, this restriction should not have a significant influence on the volume of estimated water present in the soil profile and on the rate of its fluxes (discussed in Section 3.1.1).

The simulated transpiration rates in beech forest were in general agreement with beech annual transpiration across several sites in Germany reported by Schipka et al. (2005). The average seasonal actual evapotranspiration in the spruce forest (ranging from 259 to 441 mm per vegetation season) corresponded roughly to the values of 269–351 mm reported by Šípek and Tesar (2017). They estimated AET in the same area but for a different time period and using a different soil moisture model. The differences among particular sites, namely in the rate of drainage, were given particularly by the model set-up concerning RWU parameters and soil hydraulic properties. This is based on several side-experiments when several combinations of SHP and RWU parameters from different sites were investigated. However, due to the above mentioned uncertainties in parameter determination, they are not included in the current study.

5. Conclusion

In this study, we found that the grassland and meadow sites had generally higher pressure heads compared to the forested sites. The soil water model results showed that it was caused by higher evapotranspiration rates in beech forest (by 60 mm on average) and higher drainage in spruce forest (by 35 mm on average). The differences in pressure heads between conifer forest (*Picea abies* (L.)) and deciduous forest (*Fagus sylvatica* L.) were season dependent. In dry periods (2003 and 2004), the soil with lower pressure heads was found in beech forest. Compared to conifer forest, lower pressure head values were observed namely at the bottom of the inspected soil column (down to 100 cm). This was probably caused by the more economical water consumption of coniferous trees in dry periods compared to that of broad-leaves trees, different vertical root distribution, and different soil properties. The higher transpiration of the beech forest site was proved by the model results. The beech forest evapotranspiration was on average higher by 68 mm, which was only partially compensated by higher drainage in the spruce forest, which resulted in slightly higher pressure heads. In spruce forest, the pressure heads were slightly lower (compared to beech forest) only in seasons with abundant precipitation

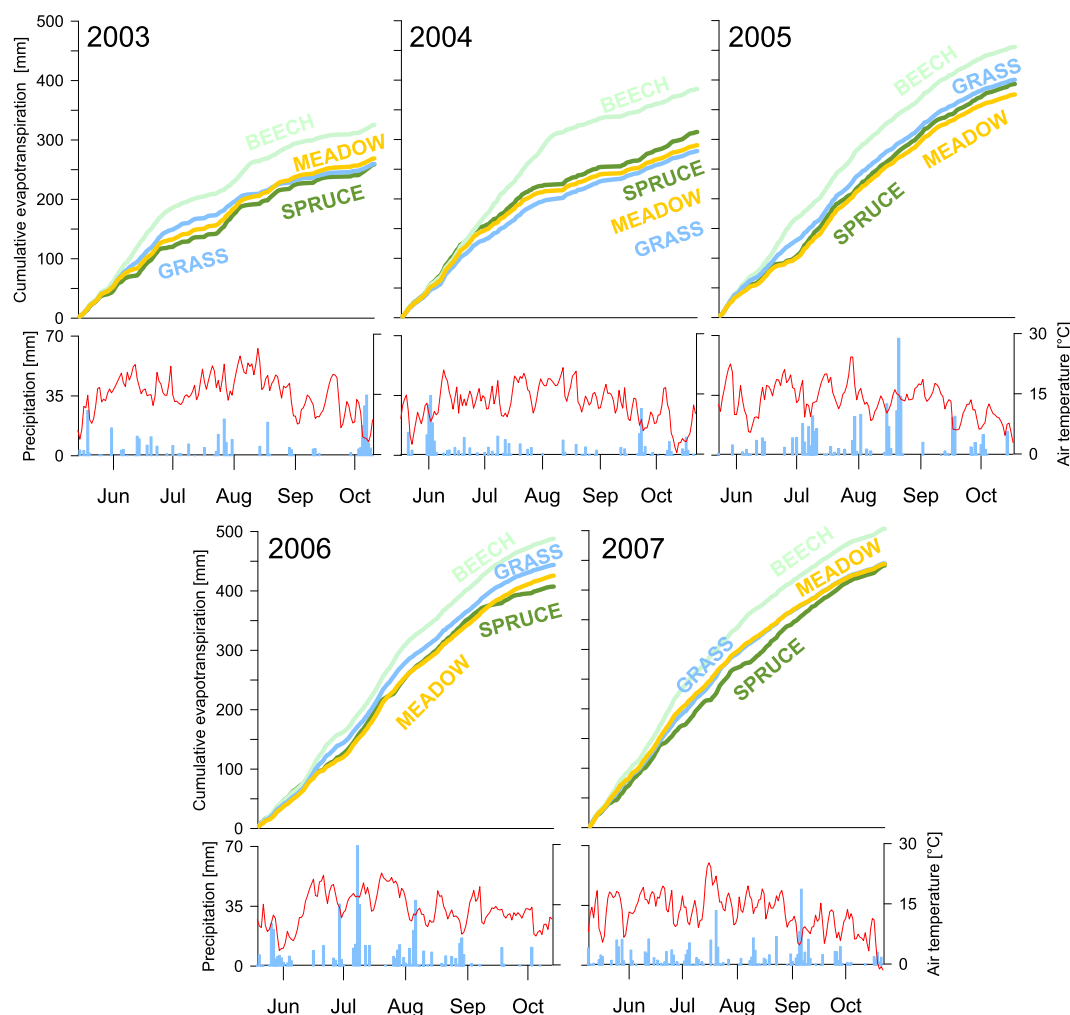


Fig. 7. Cumulative sum of daily actual evapotranspiration (interception loss, evaporation from soil surface and transpiration of vegetation) obtained from the HYDRUS-1D model.

(2005–2007), which was caused by the higher interception rate of conifer trees compared to that of deciduous trees. However, in the seasons with high precipitation, the differences among all evaluated land cover types were insignificant.

European beech (*Fagus sylvatica* L.) and Norway spruce (*Picea abies* (L.)) are two of the most dominant tree species in Central Europe. Hence, the results of this study may be transferrable to a wider region, although they might be soil type- and climate-specific.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jhydrol.2019.124285>.

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